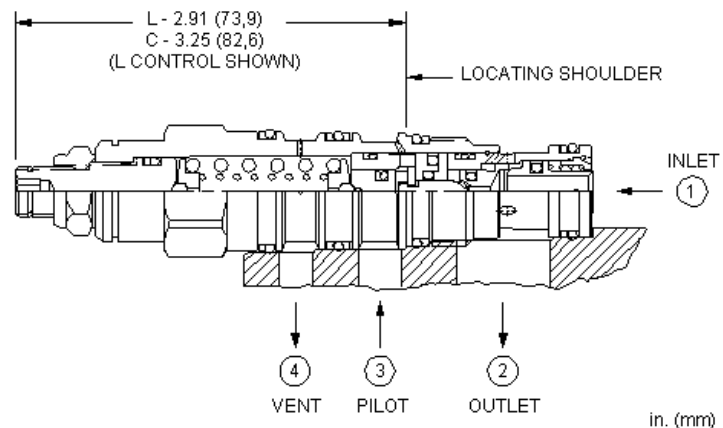
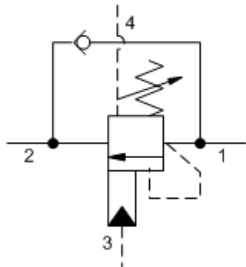


MODEL
CWCA

3:1 pilot ratio, vented counterbalance valve
CAPACITY: 60 L/min. | CAVITY: T-21A



CONFIGURATION

L	Control	Standard Screw Adjustment
H	Functional Setting Range	1000 - 4000 psi (70 - 280 bar), 3000 psi (210 bar) Standard Setting
N	Seal Material	Buna-N
(none)	Material/Coating	Standard Material/Coating

Vented counterbalance valves with pilot assist are meant to control an overrunning load. The check valve allows free flow from the directional valve (port 2) to the load (port 1) while a direct-acting, pilot-assisted relief valve controls flow from port 1 to port 2. Pilot assist at port 3 lowers the effective setting of the relief valve at a rate determined by the pilot ratio. Backpressure at port 2 does not affect the valve setting because the spring chamber references the vent (port 4).

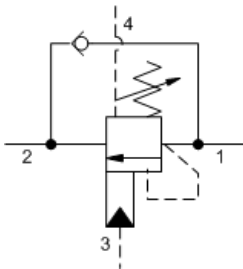
Other names for this valve include motion control valve and over center valve.

TECHNICAL DATA

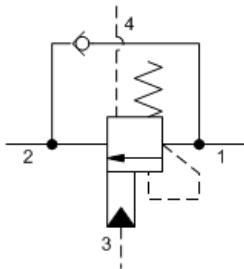
Cavity	T-21A
Series	1
Capacity	60 L/min.
Pilot Ratio	3:1
Maximum Recommended Load Pressure at Maximum Setting	215 bar
Maximum Setting	280 bar
Pilot Configuration	Sealed
Factory Pressure Settings Established at	30 cc/min.
Maximum Valve Leakage at Reseat	0,3 cc/min.
Check Cracking Pressure	2,8 bar
Adjustment - Number of Counterclockwise Turns to Increase Setting	5
Operating Characteristic	Standard
Reseat	>85% of setting
Valve Hex Size	22,2 mm
Valve Installation Torque	41 - 47 Nm
Adjustment Screw Internal Hex Size	4 mm
Locknut Hex Size	15 mm
Locknut Torque	9 - 10 Nm
Seal kit - Cartridge	Buna: 990-021-007
Seal kit - Cartridge	Polyurethane: 990-021-002
Seal kit - Cartridge	Viton: 990-021-006
Model Weight	0.19 kg.

SYMBOLS

4-Port Vented



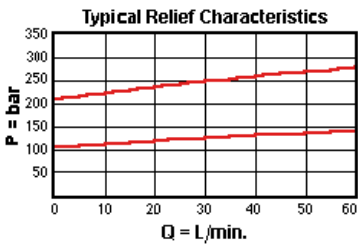
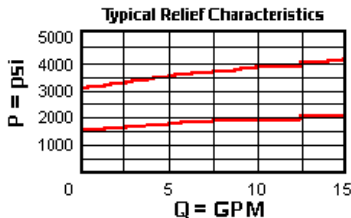
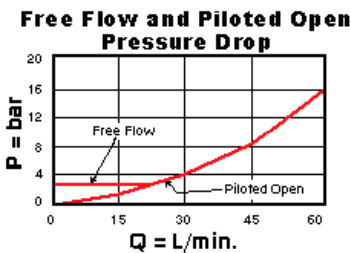
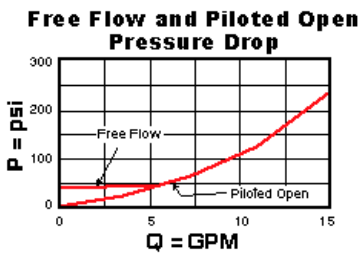
4-Port Vented, Non-adjustable



TECHNICAL FEATURES

- Counterbalance valves should be set at least 1.3 times the maximum load induced pressure.
- Turn adjustment clockwise to decrease setting and release load.
- Full clockwise setting is 200 psi (14 bar).
- All 4-port counterbalance, load control, and pilot-to-open check cartridges are physically interchangeable (i.e. same flow path, same cavity for a given frame size).
- Reseat exceeds 85% of set pressure when the valve is standard set. Settings lower than the standard set pressure may result in lower reseat percentages.
- Pressure at port 4 is added to the effective relief setting at a rate of 1 plus the pilot ratio times the pressure.
- Sun counterbalance cartridges can be installed directly into a cavity machined in an actuator housing for added protection and improved stiffness in the circuit.
- This valve has positive seals between all ports.
- With vented valves, a lower pilot ratio may be required to achieve machine stability compared to non-vented valves.
- Incorporates the Sun floating style construction to minimize the possibility of internal parts binding due to excessive installation torque and/or cavity/cartridge machining variations.

PERFORMANCE CURVES



CONFIGURATION OPTIONS

CONTROL		
Preferred Options	L	Standard Screw Adjustment
Standard Options	C	Tamper Resistant - Factory Set
FUNCTIONAL SETTING RANGE		
Preferred Options	H	1000 - 4000 psi (70 - 280 bar), 3000 psi (210 bar) Standard Setting
Standard Options	I	400 - 1500 psi (28 - 105 bar), 1000 psi (70 bar) Standard Setting

	SEAL MATERIAL	
Preferred Options	N	Buna-N
Standard Options	V	Viton
	MATERIAL/COATING	
Standard Options	/AP	Standard Material/Coating
		Stainless Steel, Passivated
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